Graduate Research Statement

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1 Introduction

My research focuses on branching and interacting particle systems across multiple scales. I am particularly interested in branching systems with genealogical dependence.

In the past, I have worked on domain applications within cosmic-ray and astroparticle physics. I am now focused on extending my background in branching processes, trees, transport and interaction to new problems.

2 Thesis Topic Areas

In this section, I describe several applications within my graduate research. High-energy astrophysics is primarily a study of relativistic particle acceleration within turbulent or shocked, magnetized fluids. The interaction of these evolving particle populations results in escaping, observable products. My work centered on disk-jet systems.

2.1 Background: Multiple Scales in Disk-Jet Systems

2.1.1 The Disk-Jet System

Disk-jet systems consist of a central compact object of some spin, surrounded by a rotating accretion disk. The hot, highly conducting accretion plasma and magnetic field are tightly coupled, so that fluid perturbations advect field lines while magnetic stresses in turn influence the fluid. As the system rotates, vertical plasma flows are wound, and magnetic tension allows for energy dissipation along this toroidal magnetic flux component, forming the jet [1, 2].

2.1.2 Fluid-Scale

Macroscopic models of disks and jet launching utilize general relativistic magnetohydrodynamics. The plasma is treated as an ideal, conducting relativistic fluid. Simulations capture the growth of large-scale turbulence, magnetorotational instabilities, and jet collimation [3].

2.1.3 Kinetic-Scale

The energies of individual particles composing this fluid (electrons and protons) are largely developed through collisionless processes. Magnetic reconnection, shock acceleration and turbulent cascades can produce approximate power-law particle energy spectra [4]. Simulations track the evolution of particle spatial-momentum distributions and fields as a collisionless Vlasov-Maxwell system. Coarse trajectories and spectra often admit stochastic or mean-field approximations [5].

2.2 Particle Transport and Interaction

My work in radiative transport and leptohadronic interactions drew on familiarity with both fluid and kinetic models for consistent descriptions of high-energy particle collisions within structured disk-jet environments.

Accelerated proton, electron and photon populations interact and evolve within regions of varying magnetization and density. Escaping particles from inelastic collisions act as observables, reflecting the underlying populations, acceleration processes and disk-jet structure. As these systems span ∼20 orders of magnitude in size, the modeling of high-energy observational signatures requires representation of behavior across spatial scales. *Different physical regimes require different treatment and compression of dominant interactions for meaningful synthesis.*

Expectations of high-energy and multiwavelength astrophysical phenomena are developed from simplified assumptions about the initial particle energy spectra, source size or density, expansion and magnetization. Particle species energy distributions evolve according to Boltzmann equations and are coupled through collisional interaction terms. The energy distribution of a particle species follows,

$$\partial_t n(E,t) = -\partial_E \left[\dot{E}(E,t) n(E,t) \right] - \alpha(E,t) n(E,t) + Q(E,t). \tag{1}$$

Here, $n(E,t)$ is the time-dependent number density for particles of energy, E. $\dot{E}(E,t)$ represents systematic energy losses such as radiative cooling and adiabatic expansion. $\alpha(E,t)$ represents losses through escape and interaction, and $Q(E,t)$ represents gains through interaction and external injection. The result is a prediction for observable multiwavelength and neutrino energy spectra [6].

In past decades, models have focused on 'steady-state' emission from a single evolved, spherical zone. My graduate work focused in part on developing more realistic models of extended, multi-zone jet structure and evolution to describe time-variable neutrino and multiwavelength observations [7]. This work with Xavier Rodrigues, "*An Evolving Leptonic Jet Model for Delayed Radio Flares in Neutrino Blazars*", is currently in preparation for submission. This experience motivates my interest in the representation of interaction phenomena across scales.

2.3 Frequentist Methods in Data Analysis

My graduate research included development and application of frequentist methods for statistical inference in multi-dimensional Poisson point processes. The IceCube Neutrino Observatory detects a mixture of astrophysical neutrino events and background cosmic rays. A parent cosmic ray interacts in the upper atmosphere, triggering a series of branching processes and a collimated shower of lower-energy daughter particles. Events are reconstructed, yielding a spatial direction, time, an energy proxy, and other observables for event topology, on a product space of observational coordinates.

The intensity may be decomposed into background and signal components,

$$\lambda(x) = \lambda_b(x) + \mu \lambda_s(x;\theta). \tag{2}$$

2.3.1 Diffuse Studies

An event excess is observed in the space of energy, declination and event topology, but is not well-resolved to known astrophysical objects. It is common to study binned event rates under the hypothesis of a 'diffuse', spatially-isotropic astrophysical flux. At the start of my graduate program, I developed sensitivity forecasts for such a study with a proposed instrumentation upgrade. This result was published [8]. I then became interested in how a unique topological detector signature from a rare class of heavy-meson-induced cosmic-ray branches could be incorporated to improve general study sensitivity. Over the course of two years, I developed and produced new simulations to characterize this observable, but ultimately found that potential inclusion would only marginally impact distinguishability.

2.3.2 Point-Source Studies

It is common to test the likelihood of observed events from a proposed set of astrophysical source locations. I was especially interested in potential neutrino emission from time-variable, jetted sources – extragalactic active galactic nuclei and galactic X-ray binary systems. I developed likelihood tests that incorporated varying temporal density in $\lambda_s(x;\theta)$. I have three works reflecting custom implementation and refinement of this likelihood, [9], [10], and *"A Time-Dependent Search for Neutrino Emission from Flaring X-ray Binaries"* – in collaboration review for submission to the Astrophysical Journal.

2.3.3 Comment

My thesis will include a section, "Statistical Inference for a Point Process with High-Dimensional, Observational Data". *This discussion will interpret improvements in sensitivity from additional observables as refinements of the underlying σ-algebra, increasing the distinguishability between component measures. This was a core concept in my applied graduate work.*

2.4 Network and Tree-Based Inference

2.4.1 Network Models for Detector Event Inference

In the early years of my graduate program, I also explored whether generative network models could improve upon table-based event reconstruction for IceCube. The detector is a lattice of ~5000 photosensors embedded within a cubic kilometer of Antarctic ice. Event photon propagation is impacted by spatial and direction-dependent scattering and absorption. As direct comparison to simulations within the full anisotropic ice model is computationally prohibited, event expectations are based on pre-simulated, tabulated photon intensities. This introduces model mismatch in the event expectation owing to interpolation. Additionally, tables are large and slow to reference.

Network architectures provide an alternative for event expectation inference and storage. An asymmetric N-Gaussian mixture model can be used to represent the photon time series expected at each sensor given a collection of event features [11]. Trained neural networks learn task-relevant data representation, offering a potential, idealized storage structure. I studied how network structure, depth, width and learning hyperparameters affect inference behavior and representational capacity. Both locally connected and convolutional layers and filters were used to capture spatially heterogeneous and shared structure in our photosensor event data. *My focus was on identifying architectures that* *offered sufficiently expressive, nondegenerate representations for our use case, as well how representation quality depends on dynamic learning.*

2.4.2 Event Classification with Gradient Boosted Decision Trees

At the start of my program, I also worked extensively with gradient boosted decision trees (GBDTs) to classify event topologies based on reconstructed observables. This project involved designing and implementing a classification pipeline for new studies, which I documented in a brief report.

3 Current Projects and Future Work

Generalizing my work, I have become interested in branching processes coupled to transport or other nonlinear interactions across multiple scales. Recently, I have focused on resource redistribution on evolving genealogical networks. I study models in which spatially or genealogically close kin are preferentially supported with resource excess, representing a local interaction rule coupling kin and resource populations. Reproduction becomes dependent on genealogy. More broadly, I am interested in dynamic branching systems and tree valued-processes.

3.1 Redistribution in Spatially-Structured Kinship Networks

In the past months, I explored this idealized system through numerical simulation and network analysis. In my work, agents receive randomly distributed resources and solicit additional resources from connected individuals when below a minimum, with each agent contributing a fraction of their excess based on relatedness.

I first analyzed a fully-connected population and hub-and-spoke network with uniform redistribution. I determined mean-field expectations as well as finite-size corrections. I then developed a spatially-limited population model with diffusion, local pairing, reproduction and mortality. Redistribution was introduced as a function of relatedness (generational distance through most recent common ancestor) and spatial proximity. Population persistence becomes possible for lower resource densities, but leads to increased network heterogeneity and centralized kin clusters. I interpreted this as an Allee-like feedback between the kinship network density, redistribution opportunity, survival, and reproduction. While this stylized work was largely exploratory and not rigorous, I have described these results in a manuscript under review at the Journal of Theoretical Biology [12].

3.2 Outlook

My current work centers on branching processes with offspring distributions dependent on genealogical structure.

4 Summary

My graduate work has centered on particle systems at multiple scales, modeling and statistical analysis, branching processes, and networks. Through applications in cosmic-ray and astroparticle physics, I developed broad exposure to diverse physical systems as well as computational methods. Over time, I have found that I am most interested in the formal structures underlying these models. I am motivated by the opportunity to apply my existing intuition to a broader set of problems at the intersection of probability, interacting particle systems, and dynamics, and to further develop the analytical tools required to understand their behavior.

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